DEPARTMENT OF MATHEMATICS & COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
(An Institution of National Importance under Ministry of Education, Govt. of India)

END SEMESTER EXAMINATIONS
M&C III Semester
MAPC-201: Linear Algebra & Applications

Date: 08-12-2025 **Time:** 09:30 AM to 12:30 PM **Max. Marks:** 50

N.B.: Answer **ALL** questions.
Answers to all parts of each question should be in one place.

Q1	Find the basis and dimension for the given subspace of $V(\mathbb{F}) = M_n(\mathbb{R})$. a) $W_1 = \{A \in V \mid A^T = A, Tr(A) = 0\}$ b) $W_2 = \{A \in V \mid A^T = -A, Tr(A) = 0\}$ c) $W_3 = \{A \in V \mid A^T = A, A^T = -A, Tr(A) = 0\}$	[7M]
Q2	Check whether $P_n(x)(\mathbb{Z}_p)$ is a vector space or not? If yes, then find the number of elements in the $P_3(x)(\mathbb{Z}_5)$.	[7M]
Q3	a) The eigenvectors of 3×3 matrix A corresponding to the eignevalues 1, 1, 3 are $\begin{bmatrix}1\\0\\-1\end{bmatrix}, \begin{bmatrix}0\\1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}$ respectively. Find the matrix A. b) Also, determine whether matrices A^T, AA^T and $(A + A^T)$ are diagonalizable or not? Justify your answers.	[4M] [3M]
Q4	Let $V^n(\mathbb{F}) = \mathbb{R}^2(\mathbb{R})$ be a vector space, $B_1 = \{(1,1),(0,1)\}$ and $B_2 = \{(1,1),(1,0)\}$ are two ordered bases. Let $\alpha = (3,-1) \in V$ then find the co-ordinates of α with respect to both the bases. Also, determine the change of basis matrix.	[7M]
Q5	Let W be a subspace $\mathbb{R}^5(\mathbb{R})$ which is spanned by the vectors $\alpha_1 = (2,-2,3,4,-1)$, $\alpha_2 = (-1,1,2,5,2)$, $\alpha_3 = (0,0,-1,-2,3)$, $\alpha_4 = (1,-1,2,3,0)$. Find $W^0 = \{f \mid f$ is a linear functional on $\mathbb{R}^5(\mathbb{R})$, and $f(\alpha) = 0$ for every α in $W\}$, dim (W) and dim (W^0).	[8M]
Q6	Define linear transformation. Find the linear transformation $T: \mathbb{R}^3(\mathbb{R}) \to \mathbb{R}^3(\mathbb{R})$ such that $T(1,2,3) = (3,4,0), T(1,0,0) = (0,1,0)$. Is this linear transformation T Unique?	[7M]
Q7	Let $B = \begin{bmatrix}3 & 1 & -1\\2 & 2 & -1\\2 & 2 & 0\end{bmatrix}$ be a matrix over field $\mathbb{R}$. Check whether matrix B is diagonalizable over $\mathbb{R}$? If yes, then specify the modal matrix. Is B maps every LI set to LI set?	[7M]

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